

Project: Image Super-Resolution using Machine Learning

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Course Code: CSE 445

Section: 07

Group No: 04

Week 1 (June 10 – June 16): Project Setup and Initial Image Selection

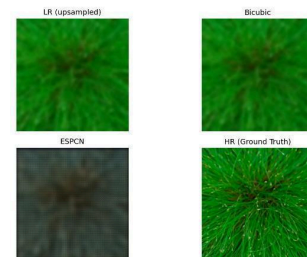
In the first week, my main objective was to establish the foundation for our project. I was responsible for selecting the first 100 images that we would use to train our super-resolution model. I also helped create the initial GitHub repository so that the team would have a central version control system for tracking all the code and data added from the beginning.

Week 2 (June 17 – June 24): Data Preprocessing and Standardization Pipeline

This week, I took full responsibility for the data preprocessing pipeline, which was essential for preparing our inputs. I wrote the `Standardize_images.ipynb` script and added 10 more images to expand the dataset. My main technical task was to write the code that standardized all of our high-resolution target images to 256 x 256 pixels. After that, I wrote a script that systematically converted these 256 x 256 images into 64 x 64 low-resolution inputs. During this time, I also handled repository operations and successfully reviewed and merged "Pull Request #1" and "Pull Request #2" into the main branch.

Week 3 (June 25 – July 4): Model Exploration and Pipeline Testing

After the datasets were prepared, I moved on to the machine learning architecture. Throughout the week, I explored super-resolution models that would work with our specific 64 x 64 to 256 x 256 pipeline. Because of its efficiency, I first attempted to implement the ESPCN architecture, but I encountered several significant challenges. The largest issue was resolving spatial dimension mismatches. To reshape the tensor back into an RGB image, ESPCN uses a PixelShuffle layer that strictly requires exactly $3 \times 4^2 = 48$ channels. Tensor shape errors during the forward pass resulted from even the smallest misconfiguration in the previous convolutional layers. Furthermore, considering our small dataset, the training process was rather unstable. After testing and discussing these problems with the team, I realized that we needed to change our approach to establish a more reliable baseline for our structural evaluations.



Week 4 (July 5 – July 11) Architecture Restructuring to SRCNN

I reorganized the model code by formally implementing the transition from the ESPCN model to the traditional SRCNN (Super-Resolution Convolutional Neural Network) architecture to improve our upscaling results. I made this switch because SRCNN directly maps a bicubic-upsampled low-resolution image to the high-resolution target. This improves our Peak Signal-to-Noise Ratio (PSNR) evaluation metric and makes the optimization process extremely stable. I successfully established the main model that we will use for our final evaluations by configuring the network to process the upsampled inputs and committing the final code ("Change ESPCN Model to SRCNN Model") directly to the main repository.

